

MILES CITY, MT, USA

WMO: 742300

Lat: 46.427N

Lon: 105.883W

Elev: 2624

StdP: 13.35

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24037

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.4	-9.1	-21.1	1.9	-14.6	-14.8	2.7	-8.7	31.2	31.4	26.9	26.8	6.9	150	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.4	98.1	65.4	94.5	64.6	91.0	64.0	69.5	88.2	68.0	86.5	66.3	84.5	10.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	98.3	75.2	62.1	91.8	74.2	60.0	85.2	73.2	35.1	87.8	33.7	86.6	32.3	83.9	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.0	23.7	20.2	DB	-23.5	104.1	7.1	3.3	-28.6	106.5	-32.8	108.4	-36.8	110.3	-42.0	112.7	(4)
(5)			WB	-23.2	73.0	6.4	2.4	-27.8	74.7	-31.6	76.1	-35.1	77.5	-39.8	79.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	20.3	23.7	34.6	46.1	55.9	66.0	75.5	73.3	62.1	47.3	33.3	22.5		
	DBStd	21.97	14.22	14.76	13.53	9.47	8.50	7.52	6.41	6.88	9.24	9.42	12.28	13.39		
	HDD50	3900	922	737	490	178	39	2	0	0	13	161	506	852		
	HDD65	7430	1387	1156	944	568	301	75	3	15	160	553	951	1317		
	CDD50	2747	0	0	11	61	220	483	791	724	376	76	5	0		
	CDD65	805	0	0	0	1	17	107	330	273	73	4	0	0		
	CDH74	10661	0	0	5	60	317	1351	4249	3486	1106	86	1	0		
	CDH80	5168	0	0	0	11	78	528	2251	1797	479	24	0	0		
	WSAvg	9.7	9.2	9.4	10.1	10.8	10.7	10.0	9.3	9.2	9.3	9.8	9.2	9.1		
	PrecAvg	13.5	0.5	0.4	0.6	1.4	2.2	2.8	1.7	1.0	1.2	0.9	0.5	0.5		
	PrecMax	20.3	1.8	1.2	1.8	4.2	6.8	6.5	6.3	3.9	4.0	6.3	2.2	1.8		
	PrecMin	5.3	0.0	0.0	0.0	0.0	0.3	0.7	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	3.9	0.4	0.3	0.4	1.1	1.4	1.5	1.5	0.9	1.0	1.0	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.0	60.2	72.5	80.9	87.1	97.0	103.1	101.4	96.3	82.8	67.7	54.1		
		MCWB	41.7	44.3	50.9	55.5	62.2	65.6	67.2	66.5	63.8	58.0	49.3	41.6		
	2%	DB	45.5	51.8	65.5	73.6	81.5	90.5	99.1	97.2	90.8	74.4	60.0	47.6		
		MCWB	37.5	40.5	47.9	53.1	59.1	65.3	66.0	64.5	61.7	53.9	45.7	38.4		
	5%	DB	41.3	45.9	58.7	68.1	77.2	86.0	95.4	93.7	85.9	68.4	54.5	42.9		
		MCWB	35.3	37.9	44.7	50.7	57.9	64.1	65.3	63.7	60.3	51.7	42.7	35.3		
	10%	DB	38.3	41.2	53.0	63.4	72.7	81.8	91.7	90.3	80.6	63.2	49.7	38.8		
		MCWB	33.4	35.2	42.0	48.1	55.7	62.3	64.9	63.3	58.2	49.2	40.5	33.1		
	0.4%	WB	42.2	45.8	51.8	57.4	65.5	71.0	72.5	70.4	67.3	58.9	50.2	42.4		
		MCDB	52.4	58.7	70.0	77.2	79.1	86.8	90.6	91.9	85.5	79.2	65.5	52.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.1	41.4	48.7	54.1	62.0	68.2	70.2	68.2	64.3	55.4	46.4	38.9		
		MCDB	44.6	50.3	63.7	71.0	76.3	84.0	89.3	88.1	82.6	70.8	58.7	47.0		
	5%	WB	35.7	38.3	45.6	51.5	59.4	66.0	68.5	66.5	62.0	52.6	43.5	35.8		
		MCDB	40.5	45.5	58.2	66.2	73.2	81.7	87.5	85.4	81.2	66.1	53.8	42.1		
	10%	WB	33.6	35.4	42.4	49.2	57.1	63.8	67.0	65.1	59.6	50.2	40.7	33.3		
		MCDB	37.9	40.7	52.2	61.7	70.2	78.9	86.0	83.8	77.7	62.1	49.0	38.2		
	5% DB	MDBR	19.0	19.8	21.9	23.2	23.0	24.1	27.4	27.8	26.7	23.2	21.4	19.1		
		MCDBR	20.9	24.7	32.0	31.9	30.3	30.6	32.1	33.0	34.8	32.6	28.6	23.1		
		MCWBR	14.6	15.5	17.2	15.6	13.7	11.6	10.4	11.2	13.6	16.0	16.5	15.2		
		MCWBR	20.1	23.3	30.6	29.2	26.8	27.1	28.1	28.1	30.1	29.5	27.8	22.5		
	5% WB	MDBR	14.7	15.0	17.0	14.9	12.8	11.7	10.6	10.6	12.3	15.0	16.4	15.5		
		MCDBR	19.0	24.7	32.0	31.9	30.3	30.6	32.1	33.0	34.8	32.6	28.6	23.1		
		MCWBR	14.6	15.5	17.2	15.6	13.7	11.6	10.4	11.2	13.6	16.0	16.5	15.2		
		MCWBR	20.1	23.3	30.6	29.2	26.8	27.1	28.1	28.1	30.1	29.5	27.8	22.5		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.258	0.273	0.308	0.332	0.345	0.357	0.360	0.320	0.285	0.259	0.247		
	taud		2.569	2.574	2.579	2.464	2.456	2.453	2.440	2.399	2.500	2.588	2.572	2.561		
	Ebn at Noon		280	299	307	302	296	292	286	281	285	284	274	269		
	Edh at Noon		19	23	26	33	34	35	35	35	29	23	18	17		
(44) All-Sky Solar Radiation	RadAvg		515	816	1188	1569	1871	2161	2294	1905	1417	907	577	409		
	RadStd		44	58	101	82	182	141	81	106	101	71	58	27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page